

Dheeraj Saraswat

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Education

Vellore Institute of Technology

Aug 23 – Jun 27

Bachelor of Technology (Computer Science) – CGPA: 8.08/10

Technical Skills

Languages: C++, SQL, Python

Fundamentals: Data Structures & Algorithms, OOPS, DBMS, Operating Systems, Computer Networks

Frameworks and Databases: RESTful API, PostgreSQL, Django, Flask, Apache Kafka

Platforms: AWS (Certified Solutions Architect – Associate), Salesforce

Data Format: JSON

Work Experience

Data Science Intern | Advance Group of Institutions, Delhi

May 26 – Jul 26

- Architected a 6-module ETL pipeline (extraction, transformation, validation, SQLite load, orchestration, scheduling) processing 3,284+ records with 4/4 automated data-quality checks passing and 0.75s execution time — reduced manual data-prep latency by ~70%
- Delivered real-time Power BI dashboards consolidating energy output, load curves and equipment health KPIs across 3+ plant datasets.
- Conducted training sessions for 10+ plant engineers and operations staff, translating forecasting model outputs and Power BI dashboards into actionable insights, improving stakeholder adoption of predictive alerts.

Projects

AgriScan: Crop Disease Detection | Python, Keras, Flask

Sep 25 – Feb 26

- Developed the yield-estimation logic converting object-tracking outputs (YOLOv4 + DeepSORT) into crop yield predictions; built supporting dashboards to visualize field-test detection accuracy and tracking reliability.
- Benchmarked the project's detection-tracking pipeline against 35+ published UAV crop-monitoring approaches through a structured literature review, informing algorithm selection.
- Resolved a localStorage quota-exceeded production bug by implementing canvas-based image compression (500KB→3KB) and a fallback retry handler, reducing storage footprint by ~99% with zero data loss.

USB Threat Monitor | WMI, python-magic, Windows Internals

Jan 25 – Apr 25

- Built a Python-based USB Threat Monitor using WMI, python-magic, and threading to detect and auto-shutdown BadUSB, Rubber Ducky, and autorun malware within 5 seconds.
- Ensured zero-tolerance data protection in high-security environments through real-time threat response.

Achievements

- Honoured with the "Best Paper Award" at the FINCON 25 International Conference for the paper titled "Proactive Fraud Prevention with AI/ML in the Digital Age." The award recognises outstanding research in financial security.
- ACM IKDD Student Travel Grant | CODS 25 — Awarded a competitive travel grant to attend the premier international conference on Data Science and Management of Data.

Licenses and Certifications

AWS Certified Solutions Architect – Associate

Aug 26 – Aug 29

Extracurricular Achievements

- Spearheaded a high-profile campus event featuring former ISRO Director Dr. Brinda V, attracting 350+ science-engaged students and fostering a strong commitment to science communication.

Position of Responsibility

Hindi Club, Vice President, VIT Bhopal University

May 24 – May 26

- Led and managed a team of 200+ members, ensuring the smooth execution of club activities.
- Organised multiple cultural events, including Hindi Diwas, Kavyanjali, Hindi Sabha, and a large-scale Holi celebration, engaging over 3,000 students in safe, inclusive, and vibrant campus activities.

Rajasthan Club, Student Coordinator, VIT Bhopal University

May 24 – Jan 25

- Led a 120+ member team, boosting club membership by 150% through new initiatives.
- Orchestrated the Rajasthan Club's inaugural and highly successful event, engaging an audience of 1000+ attendees.